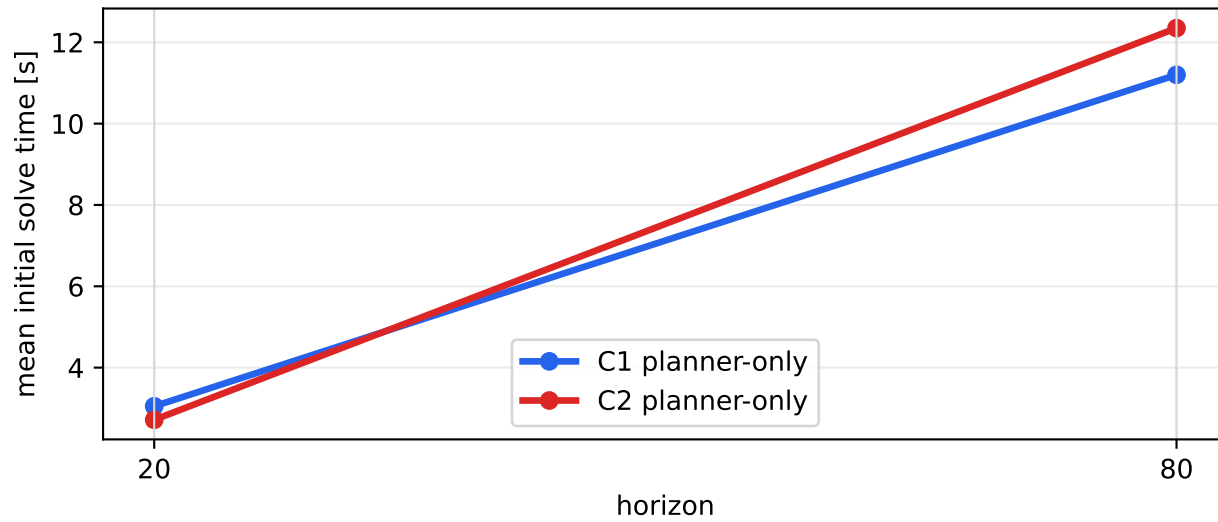
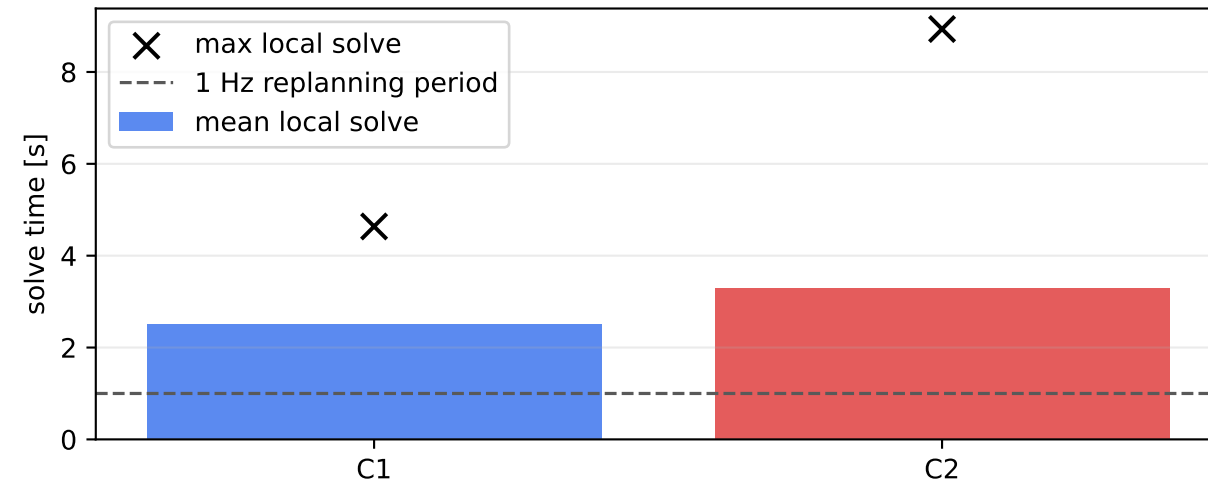


F42 - timing architecture: perception is not the only bottleneck

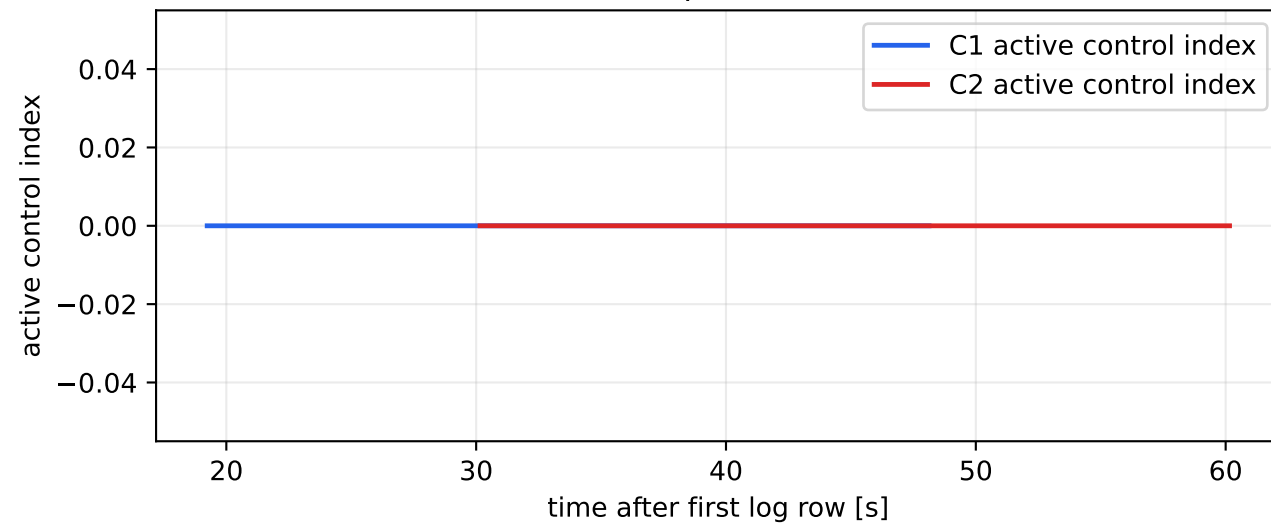
(a) Offline solve cost grows before Gazebo enters



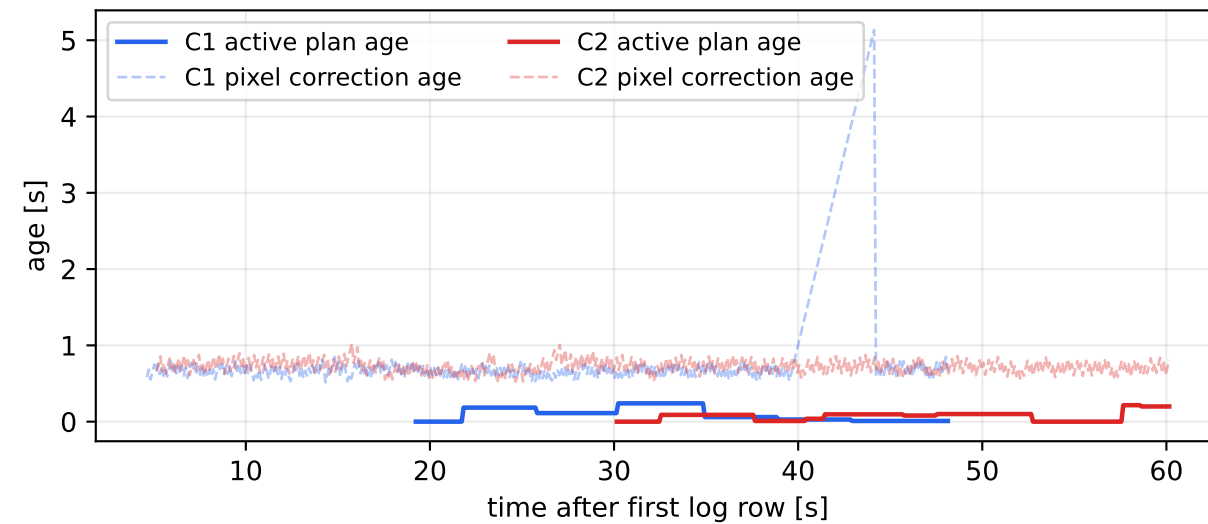
(b) Local solves are slower than the 1 Hz replan budget



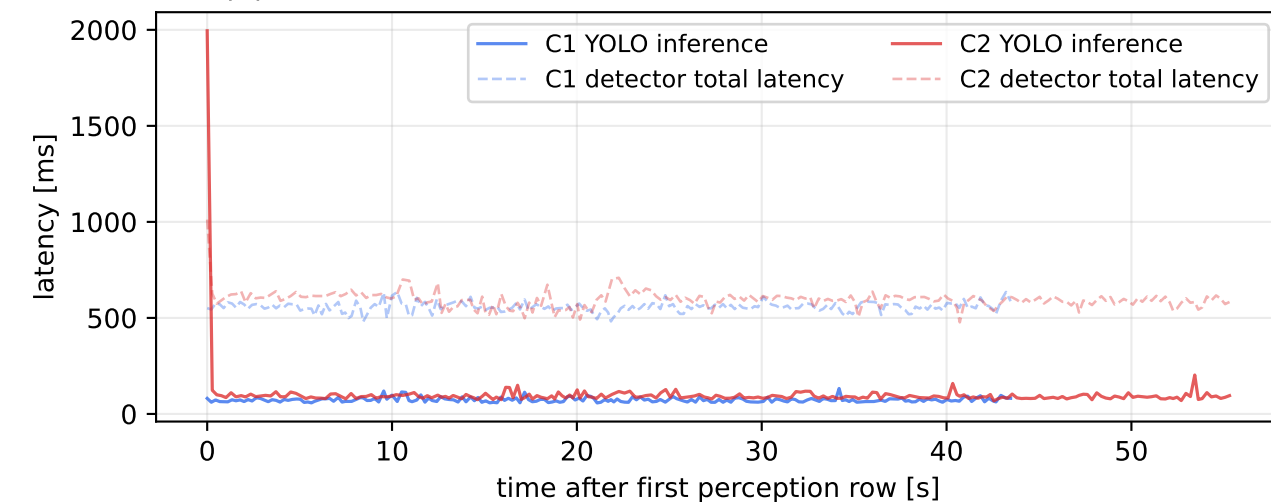
(c) Planner-result samples occur at handoff time



(d) Plans are fresh, but measurements can be stale



(e) YOLO is nontrivial, but not the dominant multi-second cost



F42 diagnosis

Gazebo outcomes:

C1: stuck, path=3.55 m, min goal=2.41 m
 local solve mean/median/max = 2510/2197/4637 ms
 active index mean/max = 0.00/0
 YOLO inference mean/max = 73/133 ms
 detector latency mean/max = 0.56/0.64 s
 C2: collision, path=4.37 m, min goal=2.50 m
 local solve mean/median/max = 3301/2761/8935 ms
 active index mean/max = 0.00/0
 YOLO inference mean/max = 101/1995 ms
 detector latency mean/max = 0.59/1.01 s

Interpretation:

- The GPU/perception path is alive, but adds ~0.6 s end-to-end latency.
- The optimizer is already multi-second offline; Gazebo makes it worse.
- F42 logs active_control_index only at planner-result publication, so it is a handoff diagnostic.
- Continuous command-tape diagnostics were added after F42 for the next run.